QUANTUMFLOOR

Computational Research Report

Job ID: 8a6980fb

1. Executive Summary

This report details the execution of the QuantumFloor diagnostic pipeline on dataset 'target'. The system evaluated multiple classical baselines against quantum algorithms to determine the optimal strategy.

2. Dataset Intelligence

Modality:	TABULAR
Total Rows:	1025
Unique Rows:	302
Duplicate Rows:	723
Total Columns:	13
Missing Values:	0.00%
Leakage Risk Detected:	0

Scientific Integrity Warnings

Warning: Duplicate rows detected: 723. GroupShuffleSplit applied to 302 unique groups to guarantee 0 cross-split leakage.

3. Data Preparation

Train/Test Split:	80/20 Stratified
Preprocessing:	Pipeline (Imputation + Scaling)
Leakage Protection:	Enforced

4. Computational Plan

Original Features:	13
Quantum Embedding Dimension:	8
Qubit Budget:	2
Classical Training Size:	816
Quantum Subset Size:	120

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5. Quantum Execution

Execution Backend: SIMULATED EXECUTION

Ansatz: AngleEmbedding + IsingZZ

Qubits: 2

Circuit Depth: 2

Total Operations: 2

6. Classical vs Quantum Baselines

Model	Type	AUROC	F1	Spec@95	Notes
full_feature_classical	Classical	0.8715	0.8205	0.6207	
logreg	Classical	0.8384	0.7831	0.2759	
svm_rbf	Classical	0.8400	0.7204	0.3017	
random_forest	Classical	0.7522	0.6138	0.2069	
qsvm	Quantum	0.8266	0.7586	0.2069	
pqk	Quantum	0.8254	0.7513	0.2414	
vqc	Quantum	0.8319	0.7299	0.2672	

7. Comparison & Explainability

Best Classical AUROC: 0.8400

Best Quantum AUROC: 0.8319

Delta: -0.0081

Explainability Status: available

PC1: 0.2408

PC2: 0.1433

8. Final Verdict

Best Overall Model: svm_rbf

Quantum Advantage: No